Finding better answers with fewer experiments

THE PROBLEM

Maximise eight unknown functions, from 2 to 8 input dimensions.

Each query costs an evaluation. We only learn how a function behaves by testing an input and observing its output.

An iterative learning loop

1 Start with provided input–output data

2 Model promising regions using a Gaussian Process

3 Propose a query using Expected Improvement and judgement

4 Observe the result, compare, and update the next decision

ONE QUERY PER FUNCTION / ROUND

Success means the best observed value, not a guaranteed global optimum.

From broad exploration to targeted refinement

Explore widely

Use initial data to learn where performance might be strong.

EARLY

Learn from feedback

Compare each round with historical bests; mix GP / EI guidance with manual checks.

MIDDLE

Tailor each function

Refine strong regions; revisit earlier bests when a new query performs worse.

LATE

Current heuristics Prioritise improvement • Explore after weak results • Avoid exact zeros • Validate formatting and dimensions

Where the evidence points

Selected changes: initial best → best observed by Week 12

F5	**1,088.86 → 8,282.12**	High values in all four inputs repeatedly worked.
F7	**1.365 → 2.436**	A recurring low-first-input region performed well.
F4	**−4.026 → 0.633**	A tight middle-input combination improved steadily.
F8	**9.598 → 9.991**	Strong results emerged from several nearby configurations.

Interpretation: promising input combinations are emerging, but dimension importance is inferred—not formally proven.

Balancing discovery with reliability

Exploit when evidence is strong

F5: the all-high setting returned 8,282.12 repeatedly. Changing its first parameter to 0.985 in Week 11 lowered the result to 8,099.25.

Decision: return to the better boundary setting.

Adapt when local moves disappoint

F6: a Week 10 query reached −0.1122, then later nearby trials performed worse.

Decision: move back toward the best historical region, while acknowledging noisy or nonlinear responses.

EXPLORE uncertain / stagnant regions

EXPLOIT repeatably successful regions

No single rule works equally well for all eight functions.

Final-round plan and what the project shows

Next experiments

- Refine near the best F2, F3, F4 and F8 observations.
- Revisit F6's Week 10 best and the strongest F7 region.
- Avoid repeating F5's identical point unless checking stability; test a purposeful nearby alternative if needed.
- Record full query history, model settings and the rationale behind each decision.

Broader ML lessons

Bayesian optimisation helps choose informative experiments when labelled observations are scarce or costly.

Gaussian Processes connect prediction with uncertainty, but local optima, limited coverage and modelling assumptions remain risks.

The useful lesson is a repeatable decision process —not just a high score.

For stakeholders: Show a simple "best result by round" chart, key successes, and the remaining uncertainty.